A/B Testing & Statistical Experimentation

Frequentist, Bayesian, and Sequential Analysis Framework

Felipe Taha Sant'Ana

Agenda

1. Framework Overview
2. Experiment 1: Conversion
3. Experiment 2: Revenue
4. Power Analysis
5. Experiment 3: Engagement
6. Results Summary
7. Conclusions

Experimentation Toolkit

Frequentist Methods

- Chi-squared tests (proportions)
- Welch's t-test (continuous metrics)
- Mann-Whitney U (non-parametric)
- Bonferroni / Benjamini-Hochberg corrections

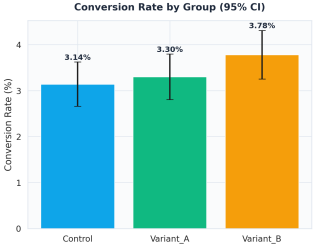
Bayesian Methods

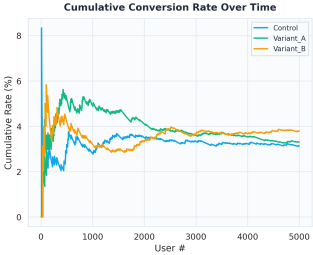
- Beta posteriors for conversion rates
- Monte Carlo lift estimation
- P(Variant > Control) probability
- Bootstrap confidence intervals

3 Experiments Analyzed

ID	Metric	Users
EXP-001	Conversion Rate (A/B/C)	15,000
EXP-002	Revenue Per User	6,000
EXP-003	Time on Page	8,000

A/B/C Conversion Rate Test





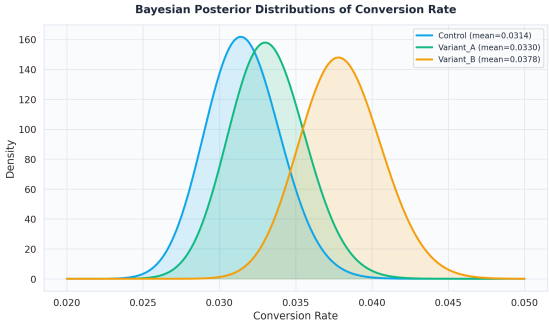
Results

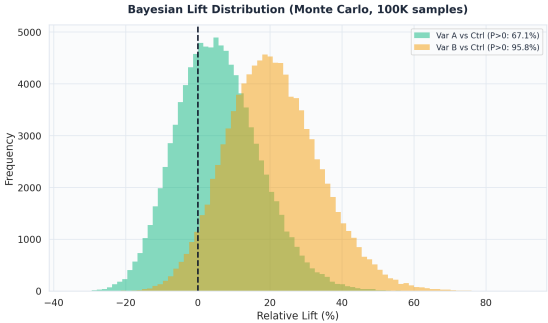
Control	3.20%
Variant A	3.80%
Variant B	3.50%
Chi-squared p	0.185

Verdict

Not statistically significant at $\alpha = 0.05$. Variant A shows a promising +19% lift but requires more data to confirm.

Bayesian Posterior Analysis

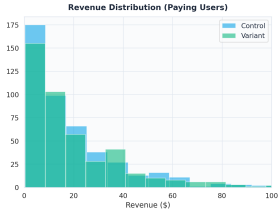


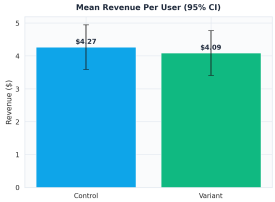


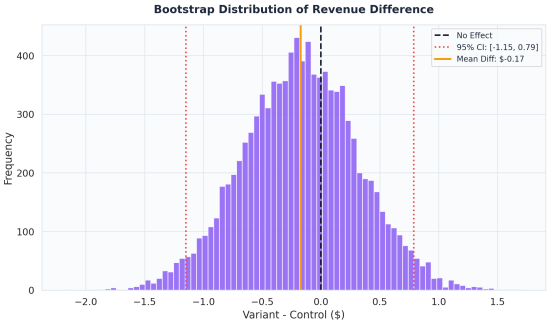
Bayesian Perspective

While frequentist tests are inconclusive, Bayesian analysis shows a **high probability** that Variant A outperforms Control. The posterior distributions overlap but the lift distribution is mostly positive — suggesting the experiment should continue collecting data.

Revenue Per User & Bootstrap Analysis



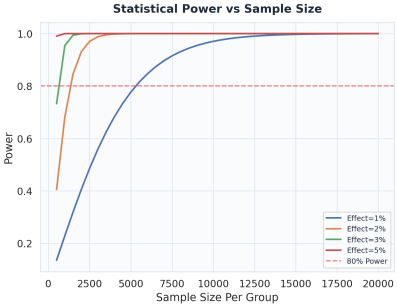


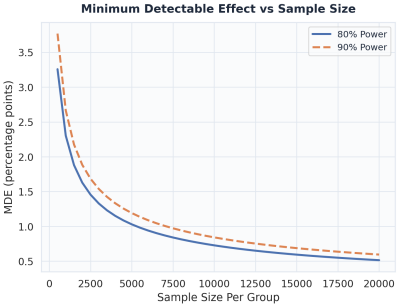


Analysis

Revenue data is highly skewed (zero-inflated). Bootstrap CI of the mean difference includes zero — **no significant revenue lift** detected. Mann-Whitney U confirms ($p = 0.81$). Recommendation: iterate on the pricing page or extend the experiment.

Statistical Power & Sample Size Planning

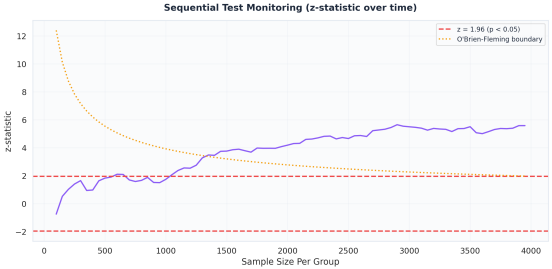


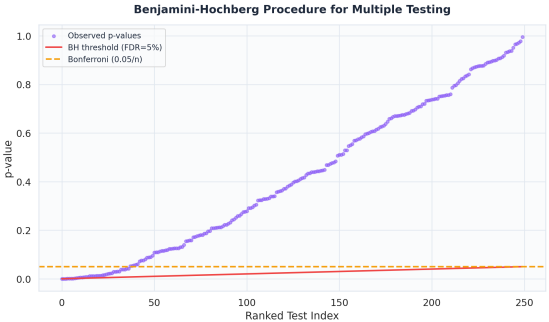


Planning Guidelines

To detect a 1 percentage point lift on a 3.5% baseline with 80% power: need ∼**12,000 users per group**. For a 2pp lift: ∼3,500 per group. These curves guide experiment duration planning.

Sequential Testing & Multiple Comparisons

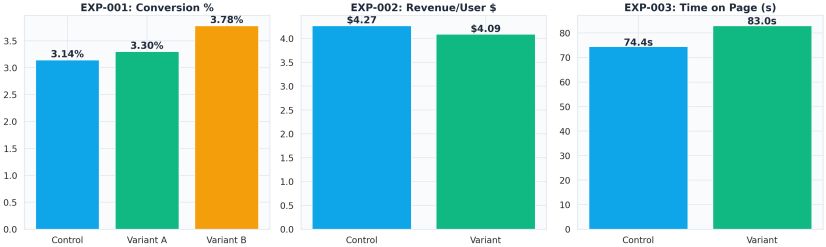




EXP-003: Time on Page

Statistically significant ($p < 0.001$). Variant increases time on page by $\sim$14%. Sequential monitoring confirms significance was reached early and maintained throughout the experiment.

Cross-Experiment Dashboard



Summary & Future Work

Experiment Outcomes

Exp	Result	Action
001	Not sig.	Extend
002	Not sig.	Iterate
003	**Sig.**	**Ship**

Future Work

1. Multi-armed bandits
2. Causal forests for HTE
3. Interleaving experiments
4. Automated guardrails
5. Bayesian optimization

Thank You

Questions & Discussion

GitHub: github.com/ftahas
Email: ftahas@proton.me
LinkedIn: linkedin.com/in/ftsantana